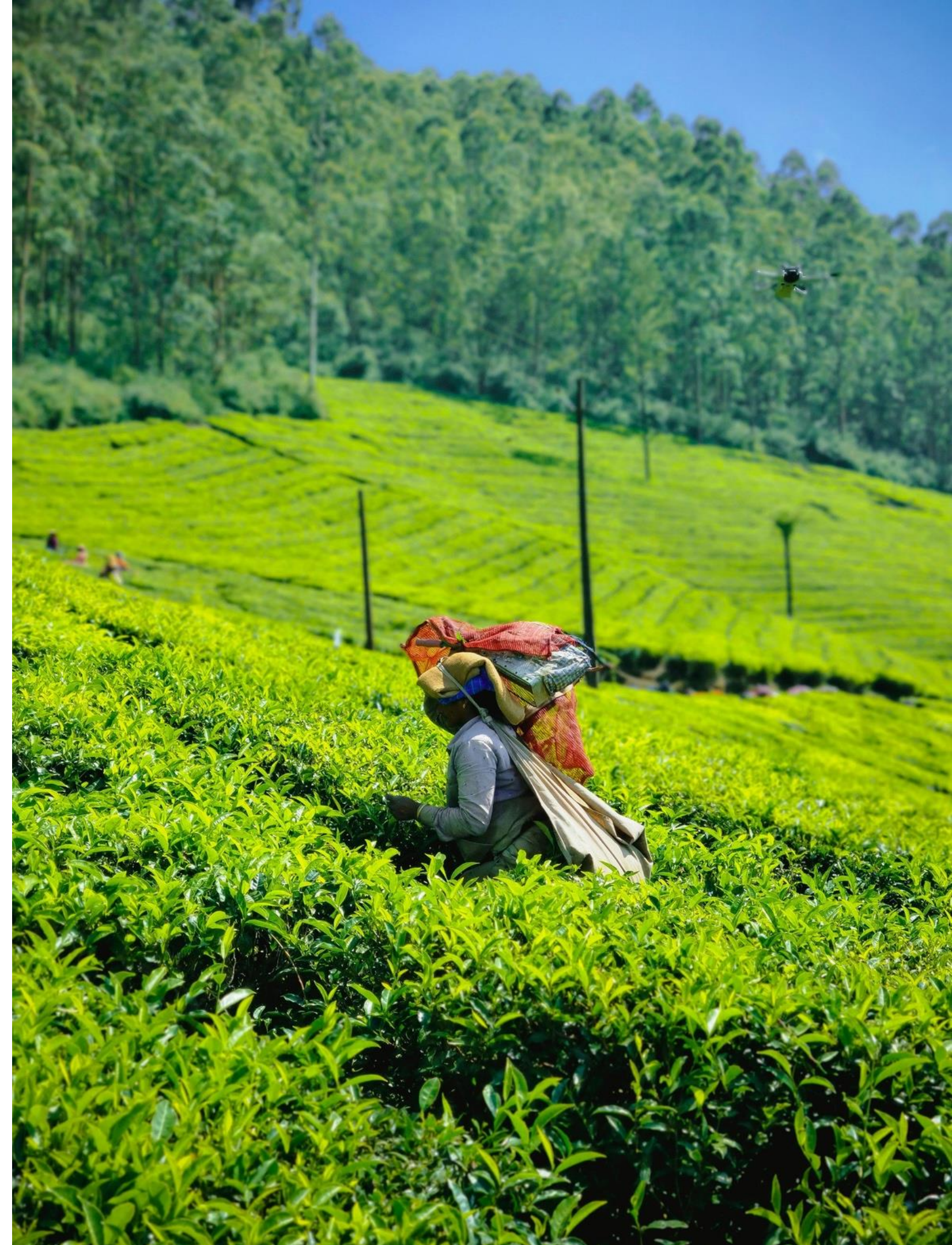


A Multi-Module AI System for End-to-End Tea Quality Assessment



Project ID: 25-26J-292





Design Excellence of the Overall Solution

What Makes the Overall Solution Well-Designed

Modular AI Architecture



- Independent AI modules for tea quality assessment
- Allows individual development, testing, and scalability

Real-Time & Data-Driven Processing



- Live camera-based image analysis
- Sensor and environmental data integration
- Instant decision support for factory supervisors

Advanced AI & Learning Mechanisms



- Computer vision for leaf, bud, texture, and defect detection
- Machine learning for grading, moisture estimation, and price prediction
- Reinforcement learning feedback loop from human tea tasters

Scalable & Industrially Practical Design



- Deployable in tea factories and processing lines
- Conveyor-based and batch-based processing support
- Scales from small plantations to large export factories

Quality, Safety & Human-Centered Support



- Assists human experts instead of replacing them
- Foreign particle detection for food safety compliance
- Improves consistency, reliability, and export quality

Modular & Scalable

Live & Integrated

Intelligent & Adaptive

Industrial & Flexible

Safe & Supportive

“Commercialization & Sustainability of TEAI – Tea Expert AI Platform”



Commercialization Model

SaaS-based AI platform for tea factories and plantations



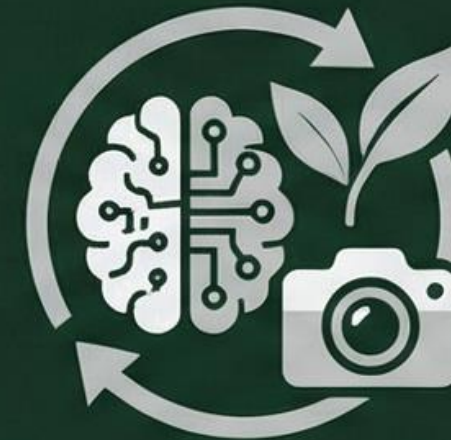
Customer Persona

Tea factory managers, quality supervisors, plantation owners



Market Opportunity

Growing demand for automation, quality consistency, and export compliance in tea industry



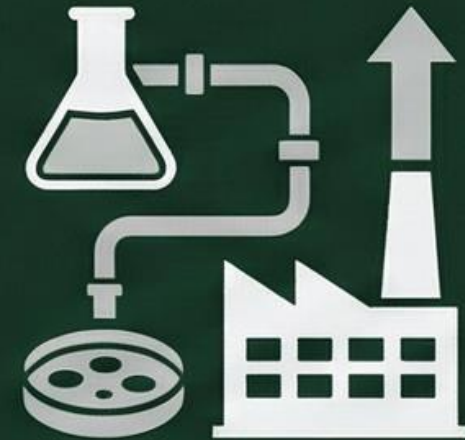
Unique Value Proposition

AI-driven visual quality assessment, moisture prediction, and expert-level decision support



Cost & Revenue Sustainability

Subscription model with scalable deployment and low per-factory operational cost



Prototype & Future Growth

Prototype validated on real tea samples, scalable to full factory integration



AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features

B.Sc. (Hons) in Information Technology specializing in Information Technology

AI-Based Computer Vision Classifier for Dried Tea Leaf Grading Using Visual Features

Problem Definition

- Tea tasting relies on subjective human visual judgement
- Expert descriptions are difficult to formalize.



Solution Approach

- Deep learning for visual feature extraction
- Colour, texture, twist and defect analysis
- Expert-in-the-loop continual learning



User Requirements Met

- Human-like descriptive outputs
- Consistent visual quality interpretation
- Adaptive calibration using expert-validated feedback.



Design Excellence / Technical Contribution

Custom Multi-Head TeaSensoryNet

- SINGLE CNN BACKBONE + 5 ATTRIBUTE HEADS (BLACKNESS, TWIST, EVENNESS, BLOOM, CLEANLINESS)

Real-Time Conveyor Capture Integration

- ARDUINO SERIAL CONTROL (AT_CAPTURE / AT_HOME) + 4K OPENCV CAPTURE → RAW DATASET CREATION

Domain-Specific Label System (0–2 Ordinal)

- TF DECODE → RESIZE → NORMALIZE (+ OPTIONAL ROI) TO ENSURE CONSISTENT PREDICTIONS

Blackness Index (BI) – Domain Feature Engineering

- DESIGNED A BI SCORE FROM LAB-L CHANNEL (DARKNESS + UNIFORMITY) TO QUANTIFY “BLACKISHNESS” LIKE A HUMAN TASTER



Automated Tea Leaf Inspection Using Computer Vision and Machine Learning



B.Sc. (Hons) in Information Technology specializing in Information Technology

Automated Tea Leaf Inspection Using Computer Vision and Machine Learning

Problem Definition

- Manual plucking quality grading is slow and inconsistent
- Difficult to evaluate large batches accurately



Solution Approach

- Vision-based bud and leaf detection
- Conveyor-based real-time analysis
- Classification: Best / Best-Below / Poor



User Requirements Met

- Batch-level quantitative reports
- Real-time factory integration
- Objective grading results



Design Excellence / Technical Contribution

Rule-Aware Computer Vision

ENCODES THE DOMAIN RULE “2 LEAVES + 1 BUD” INTO VISION-BASED CLASSIFICATION

Batch-Level Intelligence

SAMPLE-WISE PREDICTIONS AGGREGATED INTO FACTORY-LEVEL QUALITY REPORTS

Object-Level Detection

SEPARATELY DETECTS BUDS AND LEAVES INSTEAD OF TREATING THE SAMPLE AS A WHOLE

Industry-Oriented Design

BUILT TO REPLACE MANUAL INSPECTION UNDER CONVEYOR-BASED FACTORY CONDITIONS



Smart Withering Monitoring and Control System

B.Sc. (Hons) in Information Technology specializing in Information Technology

Smart Withering Monitoring and Control System

Problem Definition

- Withering decisions depend heavily on experience
- Over- or under-withering affects tea quality



Solution Approach

- Combine weight loss, images, and environment data
- Predict moisture percentage
- Estimate remaining withering time



User Requirements Met

- Real-time moisture estimation
- Actionable time recommendations
- More consistent withering control



Design Excellence / Technical Contribution

Multi-Model Comparative Analysis

COMPARED MULTIPLE MODELS TO SELECT
THE BEST.

Physics-Grounded Feature Engineering

FEATURES DERIVED FROM MOISTURE LOSS
AND PHYSICAL DRYING BEHAVIOR

Intelligent Decision Gating

CONTROLS PREDICTION FLOW BASED ON
DETECTED PROCESS STAGE

Stage-Aware Withering Time Prediction

REMAINING WITHERING TIME PREDICTED
BASED ON DETECTED PROCESS STAGE



FOREIGN PARTICLE DETECTION DURING THE FINAL PACKETING STAGE



B.Sc. (Hons) in Information Technology specializing in Information Technology

FOREIGN PARTICLE DETECTION DURING THE FINAL PACKETING STAGE

Problem Definition

- Detects small foreign particles that mistakenly mix with tea during the final conveyor-belt packaging stage.



User Requirements Met

- Provides real-time binary detection
- prevent customer complaints and product recalls.



Solution Approach

- The system successfully identifies Foreign Particles in a controlled environment



Design Excellence / Technical Contribution

Manually annotated a specialized dataset using Labelling

MANUALLY ANNOTATED OVER 90 IMAGES USING LABELIMG.

High-Speed Real-time Processing

IT PROVIDES HIGH-PRECISION DETECTION WITHOUT REQUIRING MASSIVE INDUSTRIAL SERVERS

Architectural Benchmarking

EVALUATED THREE DIFFERENT COMPUTER VISION MODELS AND SELECTED YOLOV8

Built a responsive deployment interface using Streamlit

IMPLEMENTED A FUNCTIONAL, USER-FRIENDLY UI USING STREAMLIT



TEA AUCTION PRICE PREDICTION FOR COLOMBO TEA AUCTION



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TEA AUCTION PRICE PREDICTION FOR COLOMBO TEA AUCTION

Problem Definition

- Currently Auction relies solely on the subjective experience and knowledge of tea experts.



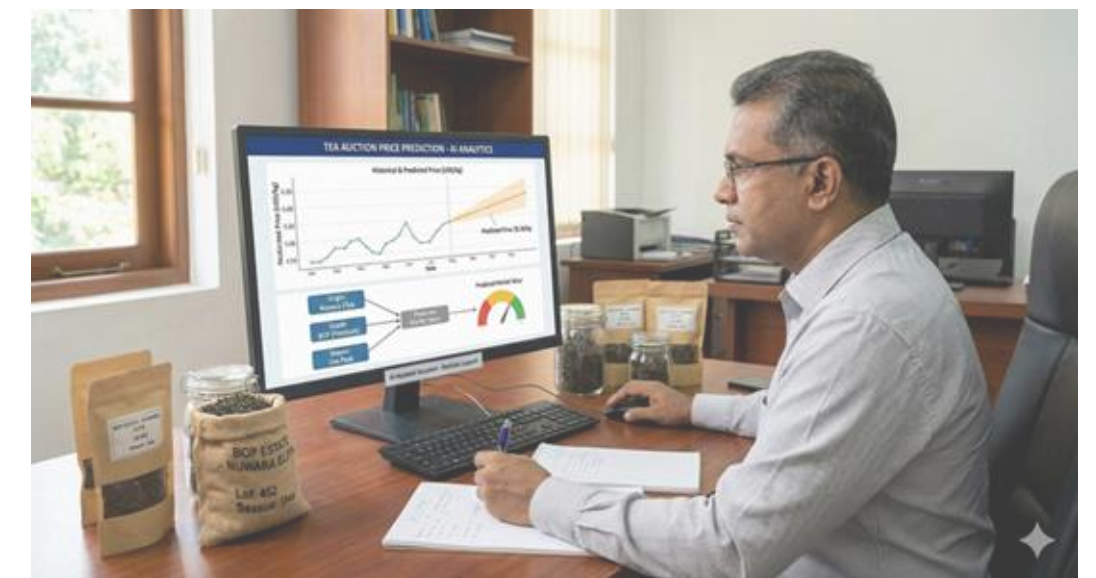
User Requirements Met

- Provides tea buyers and sellers with an estimated auction price
- Allows users to see how external variables impact the final price



Solution Approach

- replaces manual guesswork with a predictive engine
- The system predicts prices by processing five critical variables: Region, Estate, Grade, Exchange Rate and Rainfall



Design Excellence / Technical Contribution

Multi-Model

Comparative Analysis

DEVELOPED AND COMPARED 7 DIFFERENT MACHINE LEARNING MODELS TO ENSURE THE HIGHEST PREDICTION ACCURACY.

High-Performance Selection

IDENTIFIED THE "BEST MODEL" WHICH SERVES AS THE FOUNDATION FOR THE FINAL SYSTEM ARCHITECTURE

Feature Engineering for Domain Specificity

TAKE HIGHLY AFFECT FEATURES AS INPUT FEATURES

Built a responsive deployment interface using Streamlit

MPLEMENTED A FUNCTIONAL, USER-FRIENDLY UI USING STREAMLIT

Thank You



Q&A

